

LNPTM LUBRICOMPTM COMPOUND DFL22

DFL-4022

DESCRIPTION

LNP LUBRICOMP DFL22 compound is based on Polycarbonate (PC) resin containing 10% glass fiber and 10% PTFE. Added features of this grade include: Wear Resistant.

GENERAL INFORMATION	
Features	Wear resistant
Fillers	Glass Fiber
Polymer Types	Polycarbonate (PC)
Processing Techniques	Injection Molding

INDUSTRY	SUB INDUSTRY
Building and Construction	Outdoor, Lawn and Landscape
Consumer	Sport / Leisure, Personal Accessory, Home Appliances, Commercial Appliance
Electrical and Electronics	Electrical Devices and Displays, Electrical Components and Infrastructure

TYPICAL PROPERTY VALUES

Revision 20210812

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
MECHANICAL ⁽¹⁾			
Tensile Stress, yld, Type I, 5 mm/min	72	MPa	ASTM D638
Tensile Stress, brk, Type I, 5 mm/min	68	MPa	ASTM D638
Tensile Strain, yld, Type I, 5 mm/min	3.5	%	ASTM D638
Tensile Strain, brk, Type I, 5 mm/min	5.5	%	ASTM D638
Tensile Modulus, 50 mm/min	4130	MPa	ASTM D638
Flexural Modulus, 1.3 mm/min, 50 mm span	4140	MPa	ASTM D790
Tensile Stress, yield, 5 mm/min	71	MPa	ISO 527
Tensile Stress, break, 5 mm/min	67	MPa	ISO 527
Tensile Strain, yield, 5 mm/min	3.3	%	ISO 527
Tensile Strain, break, 5 mm/min	5.8	%	ISO 527
Tensile Modulus, 1 mm/min	4150	MPa	ISO 527
Flexural Stress, yield, 2 mm/min	118	MPa	ISO 178
Flexural Modulus, 2 mm/min	3900	MPa	ISO 178
IMPACT ⁽¹⁾			
Izod Impact, unnotched, 23°C	736	J/m	ASTM D4812
Izod Impact, notched, 23°C	144	J/m	ASTM D256
Instrumented Dart Impact Energy @ peak, 23°C	23	J	ASTM D3763
Multiaxial Impact	6	J	ISO 6603
Izod Impact, unnotched 80*10*4 +23°C	48	kJ/m ²	ISO 180/1U
Izod Impact, notched 80*10*4 +23°C	13	kJ/m ²	ISO 180/1A
THERMAL ⁽¹⁾			

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
HDT, 0.45 MPa, 3.2 mm, unannealed	145	°C	ASTM D648
HDT, 1.82 MPa, 3.2mm, unannealed	140	°C	ASTM D648
CTE, -40°C to 40°C, flow	5.58E-05	1/°C	ASTM E831
CTE, -40°C to 40°C, xflow	5.58E-05	1/°C	ASTM E831
CTE, -40°C to 40°C, flow	5.6E-05	1/°C	ISO 11359-2
CTE, -40°C to 40°C, xflow	5.6E-05	1/°C	ISO 11359-2
CTE, 23°C to 60°C, flow	5.6E-05	1/°C	ISO 11359-2
CTE, 23°C to 60°C, xflow	5.6E-05	1/°C	ISO 11359-2
HDT/Bf, 0.45 MPa Flatw 80*10*4 sp=64mm	146	°C	ISO 75/Bf
HDT/Af, 1.8 MPa Flatw 80*10*4 sp=64mm	140	°C	ISO 75/Af
PHYSICAL ⁽¹⁾			
Density	1.34	g/cm ³	ASTM D792
Moisture Absorption, (23°C/50% RH/24 hrs)	0.1	%	ASTM D570
Mold Shrinkage, flow, 24 hrs ⁽²⁾	0.4 – 0.6	%	ASTM D955
Mold Shrinkage, xflow, 24 hrs ⁽²⁾	0.6 – 0.8	%	ASTM D955
Mold Shrinkage, flow, 24 hrs ⁽²⁾	0.5	%	ISO 294
Mold Shrinkage, xflow, 24 hrs ⁽²⁾	0.65	%	ISO 294
Wear Factor Washer	165	10 ⁻⁴ in ³ -min/ft-lb-hr	ASTM D3702 Modified: Manual
Dynamic COF	0.37	-	ASTM D3702 Modified: Manual
Static COF	0.43	-	ASTM D3702 Modified: Manual
Density	1.33	g/cm ³	ISO 1183
FLAME CHARACTERISTICS ⁽³⁾			
UL Yellow Card Link	E207780-101344587	-	-
UL Yellow Card Link 2	E121562-101344609	-	-
UL Recognized, 94V-1 Flame Class Rating	≥3	mm	UL 94
INJECTION MOLDING ⁽⁴⁾			
Drying Temperature	120	°C	
Drying Time	4	Hrs	
Maximum Moisture Content	0.02	%	
Melt Temperature	305 – 325	°C	
Front - Zone 3 Temperature	320 – 330	°C	
Middle - Zone 2 Temperature	310 – 320	°C	
Rear - Zone 1 Temperature	295 – 305	°C	
Mold Temperature	80 – 110	°C	
Back Pressure	0.2 – 0.3	MPa	
Screw Speed	30 – 60	rpm	

(1) The information stated on Technical Datasheets should be used as indicative only for material selection purposes and not be utilized as specification or used for part or tool design.

(2) Measurements made from laboratory test coupon. Actual shrinkage may vary outside of range due to differences in processing conditions, equipment, part geometry and tool design. It is recommended that mold shrinkage studies be performed with surrogate or legacy tooling prior to cutting tools for new molded article.

(3) UL Ratings shown on the technical datasheet might not cover the full range of thicknesses and colors. For details, please see the UL Yellow Card.

(4) Injection Molding parameters are only mentioned as general guidelines. These may not apply or may need adjustment in specific situations such as low shot sizes, large part molding, thin wall molding and gas-assist molding.



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